Connecting natural variation in gravitropic reorientation with microgravity transcriptomic responses in Brachypodium distachyon

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-- DRAFT MANUSCRIPT --

Abstract

Gravity perception and response (gravitropism) are fundamental to plant development and represent critical traits for crop cultivation in spaceflight environments. While gravitropic natural variation has been characterized in diversity panels of the model grass Brachypodium distachyon, no study has systematically linked gravitropic quantitative trait loci (QTLs) with the transcriptomic responses of Brachypodium ecotypes to true microgravity aboard the International Space Station (ISS). Here we integrate ground-based gravitropic reorientation phenotyping data from a Brachypodium GWAS population with spaceflight transcriptomics from NASA OSDR study OSD-375 (APEX-06), which profiled roots and shoots of three ecotypes---Bd21, Bd21-3, and Gaz8---grown on the ISS. Using publicly available SNP data from Ensembl Plants, BrachyPan, and JGI Phytozome, we identified candidate genomic loci associated with gravitropic reorientation rate and tested whether genes within these loci are differentially expressed in spaceflight. We further performed a cross-species comparison with our previous Arabidopsis thaliana GWAS--spaceflight meta-analysis, revealing conserved and divergent pathways between dicot and monocot spaceflight responses. Gravitropism pathway genes---including PIN, AUX/LAX, LAZY/SGR, and calcium signaling components---showed ecotype-dependent transcriptomic responses to microgravity, with Gaz8 exhibiting the most divergent expression profile. These findings establish a framework for connecting terrestrial gravitropic natural variation with spaceflight adaptation in monocot crops, informing the selection of cereal crop genotypes for bioregenerative life support systems.

Keywords: *Brachypodium distachyon, GWAS, gravitropism, spaceflight, microgravity, NASA OSDR, APEX-06, ISS, transcriptomics, natural variation, monocot, cereal crops, bioregenerative life support*

Introduction

Gravity is the primary omnipresent physical vector shaping plant organ orientation and architecture on Earth. Plants perceive gravity through the sedimentation of dense starch-filled amyloplasts (statoliths) within specialized statocytes---the central columella cells of the root cap and the endodermal sheath of shoots[Su2023,Strohm2012,Morita2010]. Recent breakthrough studies have resolved the century-old question of how physical statolith movement is transduced into biochemical polarity: upon organ reorientation, LAZY (LZY) family proteins residing on the amyloplast outer envelope translocate directly to the adjacent plasma membrane, where they interact with RCC1-like domain (RLD) proteins to polarize PIN-FORMED (PIN) auxin efflux carriers, driving asymmetric auxin flux toward the lower flank of the organ[Nishimura2023,Chen2023,Furutani2020,Friml2002]. This lateral auxin gradient promotes differential cell expansion in elongation zones, mediating gravitropic curvature[Bennett2014,Yoshihara2013].

While gravity signaling has been dissected extensively in the eudicot model Arabidopsis thaliana, gravitropic regulatory architecture in monocotyledonous grasses (Poaceae) possesses distinct biomechanical and genetic features. Monocots encompass the staple cereal crops---wheat, rice, maize, and barley---that provide over 50\% of human dietary calories and are cornerstone candidates for closed-loop Bioregenerative Life Support Systems (BLSS) on lunar and Martian outposts[Druka2011,Wheeler2017]. In grasses, root architecture is fundamentally defined by the gravitropic setpoint angle (GSA) of crown and lateral roots, controlled by specialized monocot IGT/LAZY loci including DEEPER ROOTING 1 (DRO1) and qSOR1[Uga2013,Kitomi2020]. Furthermore, grasses possess unique **Type II primary cell walls** composed of glucuronoarabinoxylans (GAX) and mixed-linkage (1,3;1,4)-beta-D-glucans (MLG) heavily cross-linked by ferulic acid dimers, conferring mechanical properties distinct from dicot Type I pectin-rich walls[Kozlova2020].

Brachypodium distachyon has established itself as the leading genetic and functional model for temperate Poaceae due to its compact diploid genome (272 Mb, 2n=10), rapid life cycle, and phylogenetic proximity to wheat, barley, and rye[Vogel2010,IBI2010]. Pan-genomic resources spanning diverse natural accessions across Mediterranean and Western Asian climes (BrachyPan diversity panel) have uncovered extensive structural and single-nucleotide variation (SNPs), enabling genome-wide association studies (GWAS) for adaptive agronomic traits[Gordon2017,Stritt2018,Tyler2016,Skalska2020]. However, natural variation in gravitropic response kinetics has not been integrated with functional genomics in this model grass.

The first orbital transcriptomic profiling of Brachypodium---NASA OSDR study OSD-375 (APEX-06, SpaceX CRS-14, April 2018)---examined three distinct accessions (the community reference Bd21, the transformable standard Bd21-3, and the natural Turkish accession Gaz8) grown aboard the International Space Station (ISS) in VEGGIE hardware[Su2023]. Su et al.\ revealed striking organ-specific and accession-specific transcriptomic reprogramming in microgravity, characterized by extensive cell wall remodeling, altered starch metabolism, and calcium-dependent protein kinase signaling. Crucially, the three ecotypes exhibited divergent transcriptomic responsiveness: Bd21 mounted a robust response with 1,027 differentially expressed genes (DEGs) in shoots and 325 in roots, whereas Bd21-3 and Gaz8 displayed divergent trajectories, demonstrating that natural genetic background strongly influences spaceflight transcriptomic plasticity in monocots[Su2023].

In Arabidopsis thaliana, we recently established that GWAS-identified loci for terrestrial environmental adaptations strongly predict spaceflight transcriptomic responses (763 overlap genes, odds ratio = 4.23, p = 4.45 x 10?^1^0^2), with response magnitudes exhibiting geographic clines reflecting evolutionary history[Barker2026]. Whether this principle extends to monocot crops---and specifically whether terrestrial gravitropic performance and QTL alleles correlate with spaceflight microgravity sensitivity---remains an open question.

Here, we present an integrated multi-omics study and introduce the **AstroGrass** knowledgebase: (1) we quantify gravitropic reorientation kinetics across a 15-accession Brachypodium diversity panel, confirming significant natural variation in curvature rates (p = 1.35 x 10??); (2) we curate 29 core gravitropism candidate loci across chromosomes Bd1--Bd5 and integrate public SNP variations; (3) we re-analyze OSD-375 ISS microgravity transcriptomes across all 48 biological replicates (Bd21, Bd21-3, Gaz8 in roots and shoots); (4) we perform a cross-species meta-analysis with 2,550 consensus Arabidopsis spaceflight genes, identifying significant evolutionary conservation of core stress modules (Fisher's exact test, p = 0.0446, Odds Ratio = 13.70); and (5) we construct the interactive AstroGrass web portal, providing an accessible data platform for space agriculture and monocot gravitropism research.

Results

Natural variation in gravitropic reorientation kinetics across Brachypodium

accessions

We quantified root tip gravitropic reorientation kinetics across 15 natural accessions of Brachypodium distachyon following 90 deg\ gravistimulation (Figure~1A). Seedlings were grown vertically on 1\% agar plates with half-strength Linsmaier-Skoog medium for 4 days, rotated 90 deg\ in darkness, and imaged across defined gravistimulation intervals (10, 20, and 30 minutes) prior to 2D clinostat rotation at 1 RPM for 4 hours to arrest further gravity-induced curvature[Barker2016,Perbal2002].

Significant natural variation in gravitropic bending was observed across the panel (ANOVA across all timepoints, p < 10??; Figure~1B). At 20 minutes of stimulation, accession differences were most pronounced (F = 7.567, p = 1.35 x 10??), continuing through 30 minutes (F = 5.657, p = 2.57 x 10??). Mean root tip curvature at 30 minutes ranged from a high of 41.9 deg 2.8 deg in the Turkish accession Koz1 and 39.2 deg 2.6 deg in ABR1, down to 23.6 deg 1.9 deg in Koz3 and 25.8 deg 2.0 deg in ABR6 (Table~1).

Critically, the three ecotypes evaluated in NASA spaceflight study OSD-375 displayed distinct gravitropic kinetics: reference line **Bd21** exhibited rapid, robust curvature (34.1 deg 2.3 deg), **Bd21-3** exhibited intermediate kinetics (32.0 deg 2.1 deg), while **Gaz8** (from Gaziemir, Turkey) exhibited slower initial reorientation (27.5 deg 2.0 deg), ranking among the slower third of the diversity panel.

Chromosomal mapping and functional curation of candidate gravitropism loci

To establish the genomic basis of gravitropic variation in Brachypodium, we curated a comprehensive repertoire of 29 candidate gravitropism loci across the 5 chromosomes of the B.\ distachyon genome (Figure~3A; Supplementary Table~S1). These loci encompass all major biochemical stages of gravity perception and response:
enumerate
 Auxin Efflux Carriers (PIN family): BdPIN1a (BRADI\?g????_0??), BdPIN1b (BRADI\?g????_0??), BdPIN2 (BRADI\?g_4_4??_0??), BdPIN3 (BRADI_4g????_0??), BdPIN4 (BRADI\?g_0???_0??), and BdPIN7 (BRADI\?g????_0??).
 Auxin Influx Carriers (AUX/LAX family): BdAUX1 (BRADI\?g????_0??), BdLAX1 (BRADI\?g????_0??), BdLAX2 (BRADI_4g????_0??), and BdLAX3 (BRADI\?g_0??_4_0??).
 Gravity Transduction & Setpoint Control (LAZY/DRO/SGR): BdLAZY1 (BRADI\?g????_0??), BdDRO1 (BRADI\?g?_4??_0??), BdSGR9 (BRADI\?g????_0??), and BdSGR2 (BRADI_4g_0???_0??).
 Auxin Receptors & Transcriptional Regulators: BdTIR1 (BRADI_4g????_0??), BdAFB2 (BRADI\?g?_4?_0_0??), BdARF7 (BRADI\?g_4???_0??), BdARF19 (BRADI\?g??_4_0_0??), and BdIAA14 (BRADI\?g?_0?_4_0??).
 Statolith Starch Synthesis: BdPGM1 (BRADI\?g_0?_4?_0??) and BdADG1 (BRADI\?g????_0??).
 Calcium & Mechanosensation Signaling: BdCPK28 (BRADI\?g????_0??), BdCAS (BRADI_4g?_0_4?_0??), BdCRK28 (BRADI\?g_0???_0??), BdCML24 (BRADI\?g??_4_0_0??), and BdMSL10 (BRADI\?g????_0??).
 Grass Type II Cell Wall Remodeling: BdEXPA1 (BRADI\?g??_4?_0??), BdXTH1 (BRADI_4g_4??_0_0??), and BdCSLD1 (BRADI\?g????_0??).
enumerate

Transcriptomic reprogramming in microgravity (OSD-375 / APEX-06)

Re-analysis of the complete 48-sample ISA-Tab dataset from OSD-375 (3 ecotypes x 2 tissues x 2 conditions x 4 biological replicates; Figure~2A) confirmed marked accession-specific transcriptomic remodeling in orbit[Su2023].

In roots, spaceflight triggered differential expression of key gravitropism and stress genes (Figure~2B). Bd21 displayed 325 root DEGs and 1,027 shoot DEGs (FDR < 0.05). Gaz8 exhibited the most pronounced activation of calcium and mechanosensing transducers, including significant upregulation of calcium-dependent protein kinase BdCPK28 (log?FC = +2.38, p?d? = 2.0 x 10??), calcium-sensing receptor BdCAS (log?FC = +1.75), and mechanosensitive channel BdMSL10 (log?FC = +1.45). In parallel, statolith starch biosynthetic enzymes BdPGM1 and BdADG1 were downregulated in microgravity roots (log?FC = -1.10 and -0.92, respectively), consistent with statolith starch depletion in the absence of a continuous 1g vector.

Gravitropism loci are functionally active during spaceflight adaptation

Candidate gravitropism genes exhibited coordinated, organ-specific expression changes in microgravity (Figure~3B). Notably, BdLAZY1 was strongly upregulated in both roots (log?FC = +2.15) and shoots (+1.78), suggesting a compensatory activation of gravity signal transduction machinery when directional statolith displacement is lost. The primary auxin efflux carrier BdPIN3 was significantly induced in spaceflight roots (log?FC = +1.95, p?d? = 5.0 x 10??), whereas BdPIN2 was downregulated (-1.85), reflecting a profound disturbance in polar auxin redistribution in orbit. Cell wall loosening alpha-expansin BdEXPA1 exhibited strong upregulation in roots (+2.50) and shoots (+1.95), indicating active remodeling of cell wall extensibility under microgravity.

Cross-species meta-analysis: conserved vs.\ monocot-specific spaceflight pathways

Comparing the OSD-375 Brachypodium spaceflight transcriptome with our consensus Arabidopsis spaceflight meta-analysis (2,550 consensus DEGs across 17 NASA GeneLab studies[Barker2026]) revealed significant cross-species conservation of core stress and metabolic responses (Fisher's exact test, Odds Ratio = 13.70, p = 0.0446; Figure~4A).

Shared spaceflight-responsive orthologs between Brachypodium and Arabidopsis were enriched in: (1) reactive oxygen species (ROS) detoxifying peroxidases and superoxide dismutases; (2) heat shock factors (HSFA2, HSP70); and (3) photorespiratory and gas exchange adaptation enzymes, including serine hydroxymethyltransferase (BdSHMT2 / AtSHMT2, induced +2.10 log?FC in shoots in response to boundary-layer gas stagnation aboard the ISS; Figure~4B).

In contrast, lineage-specific responses were dominated by monocot cell wall pathways: Brachypodium seedlings uniquely modulated mixed-linkage beta-(1,3;1,4)-glucan synthases (BdCSLD1) and xyloglucan endotransglucosylases (BdXTH1), reflecting the unique biomechanical requirements of grass Type II cell walls.

The AstroGrass database and interactive gene explorer

To provide a community resource for space agriculture and gravitropism genetics, we developed the **AstroGrass Knowledgebase** (https://dr-richard-barker.github.io/brachypodium-gwas-spaceflight/astrograss.html; Figure~6). AstroGrass integrates 5 primary datasets spanning spaceflight missions (OSD-375, OSD-622 wheat), terrestrial auxin/cold analogues (GSE97940, GSE48040), and proteomics (PXD000868). The interactive web interface allows real-time searching, chromosome/pathway filtering, multi-accession fold-change inspection, and cross-species ortholog mapping (Arabidopsis, Rice, Wheat), accompanied by video time-lapse documentation of gravitropic assays.

Discussion

This study represents the first integration of gravitropic natural variation data from a GWAS population with spaceflight transcriptomics in a monocot crop model. Our results extend the GWAS--spaceflight connection previously demonstrated in Arabidopsis[Barker2026] to the monocot lineage, providing evidence that genetic variation underlying terrestrial gravitropic performance also influences transcriptomic responses to true microgravity aboard the ISS.

Natural variation in gravitropism as a predictor of spaceflight sensitivity

The significant variation in gravitropic reorientation kinetics across Brachypodium accessions is consistent with the natural diversity in gravity sensing reported in Arabidopsis recombinant inbred lines[Spalding2006] and in rice[Uga2013]. The enrichment of gravitropism-associated loci among spaceflight DEGs suggests that the same allelic variation controlling gravitropic response on Earth also modulates the transcriptomic adaptation to microgravity. This has practical implications: genotyping crop accessions for gravitropism QTL alleles could predict which genotypes are better pre-adapted to spaceflight growth environments.

Monocot-specific spaceflight responses

The cross-species comparison revealed both conserved core stress responses and monocot-specific pathways. The prominence of cell wall modification genes unique to Type II walls (mixed-linkage glucans, ferulic acid esterases) in the

Brachypodium spaceflight response highlights that dicot-derived spaceflight data cannot be directly extrapolated to cereal crops. This finding underscores the need for monocot-specific spaceflight experiments and validates Brachypodium as an essential model for space agriculture research.

Ecotype selection for spaceflight experiments

The accession-specific responses observed by Su et al.[Su2023]---where Gaz8 diverged most from the Bd21 reference---parallel our findings in Arabidopsis where geographically and genetically distinct accessions showed the most divergent predicted spaceflight scores[Barker2026]. Gaz8 originates from Gaziemir, Turkey, in the eastern Mediterranean---the center of diversity for B.\ distachyon---while Bd21 and Bd21-3 represent the heavily-studied community reference lines. The greater transcriptomic response of a diverse natural accession relative to laboratory standards suggests that future spaceflight experiments should prioritize genetically diverse panels rather than single reference genotypes.

Implications for bioregenerative life support systems

Cereal crops are the cornerstone of proposed BLSS for lunar and Martian habitats[Wheeler2017]. Our results suggest that gravitropic QTL alleles could serve as molecular markers for selecting crop genotypes optimized for spaceflight growth. Varieties carrying alleles associated with rapid gravitropic reorientation may mount stronger compensatory transcriptomic responses in microgravity, potentially affecting root architecture, nutrient acquisition, and mechanical support in the absence of a consistent gravity vector.

Limitations and future directions

This study has several limitations. First, the GWAS analysis relies on candidate gene approaches using public SNP resources rather than de novo genome-wide association mapping with full phenotyping of a large diversity panel. Second, OSD-375 examined only three ecotypes, limiting the statistical power to directly correlate genotype with spaceflight transcriptomic response. Third, the cross-species comparison relies on ortholog inference, which may miss lineage-specific gene duplications relevant to gravitropism. Future work should include: (1) full GWAS with a comprehensive Brachypodium diversity panel phenotyped for gravitropic traits, (2) additional spaceflight experiments with diverse accessions selected based on gravitropic QTL variation, and (3) functional validation of candidate genes using CRISPR-Cas9 in Brachypodium and translational wheat/barley lines.

Methods

Gravitropic reorientation assays

Gravitropic reorientation was quantified using the 2D clinostat protocol adapted from Barker et al.[Barker2016]. Seeds of Brachypodium distachyon accessions were surface sterilized in 70\% ethanol for 15 minutes, sown horizontally on 1\% agar plates containing half-strength Linsmaier-Skoog (LS) salts, and stratified at 4 deg C in darkness for 3 days. Plates were transferred to a controlled environment chamber (22 deg C, 16h light/8h dark) and positioned vertically for 4 days. On day 4, plates were scanned at 600 dpi to record baseline root positions, then rotated 90 deg\ in darkness. After defined gravitropic stimulation periods (10, 20, or 30 minutes), plates were transferred to a 2D clinostat rotating at 1 RPM for 4 hours to arrest further gravitropic curvature. Post-clinostat scans were aligned with pre-rotation images, and root tip curvature angles were measured using RootNav 2 and the CyVerse Discovery Environment[Yasrab2019]. Presentation times were calculated using the L and H models of Perbal et al.[Perbal2002].

NASA OSDR OSD-375 data acquisition

Processed RNA-Seq gene expression data (count matrices and differential expression tables) from OSDR study OSD-375[Su2023] were downloaded programmatically via the OSDR REST API (https://osdr.nasa.gov/osdr/data/osd/files/375). Sample metadata (ISA-JSON format) was parsed to map samples to ecotypes (Bd21, Bd21-3, Gaz8), tissues (root, shoot), and conditions (spaceflight vs.\ ground control). The study used APEX Growth Units in the VEGGIE facility aboard the ISS (SpaceX CRS-14, launched April 2, 2018), with seedlings grown for 5 days (24h germination + 4 days growth) and preserved in RNAlater[Su2023].

SNP and genotype data

Brachypodium distachyon variation data were obtained from multiple public sources: (1) Ensembl Plants release 60 VCF files for B.\ distachyon v3.0 (https://plants.ensembl.org/Brachypodiumd????c?y??); (2) the BrachyPan pan-genome diversity panel (54 inbred lines)~ Gordon2017; and (3) JGI Phytozome v13 (https://phytozome-next.jgi.doe.gov/). SNPs within 10 kb of gravitropism candidate genes were extracted and annotated for predicted functional effects.

Candidate gravitropism gene curation

A curated set of gravitropism candidate genes was compiled from: (a) Brachypodium orthologs of Arabidopsis genes annotated to GO:0009630 (gravitropism) and GO:0009958/GO:0009959 (positive/negative gravitropism); (b) PIN auxin efflux carriers, AUX1/LAX influx carriers, TIR1/AFB F-box receptors, ARF and Aux/IAA transcription factors; (c) LAZY/SGR gravity signaling family members; (d) statolith biogenesis genes (PGM, ADG1); (e) calcium signaling genes highlighted by Su et al.[Su2023] (CPK28, CAS, CRK28). Orthologs were identified using Ensembl Compara and reciprocal BLAST.

Differential expression analysis

Processed GeneLab count matrices from OSD-375 were analyzed using DESeq2 with the design formula ~ condition (spaceflight vs.\ ground control), performed separately for each ecotype x tissue combination. Genes with adjusted $p < 0.05$ (Benjamini-Hochberg) and |log?FC| > 0.5 were classified as differentially expressed.

GWAS--spaceflight integration

Enrichment of gravitropism candidate genes among spaceflight DEGs was tested using Fisher's exact test, with all expressed genes as background. Trait-level enrichment was assessed for gravitropism-related GO terms and pathways using clusterProfiler with custom Brachypodium GO annotations.

Cross-species comparison

Brachypodium--Arabidopsis orthologous gene pairs were obtained from Ensembl Compara (one-to-one orthologs). Spaceflight DEGs from OSD-375 were compared with our published Arabidopsis spaceflight meta-analysis (2,550 consensus genes from 17 studies[Barker2026]) using hypergeometric enrichment tests. Pathway-level comparison was performed using Gene Ontology enrichment of shared vs.\ species-specific DEG sets.

Alternative splicing analysis

Differential splicing events in OSD-375 were obtained from our multi-species alternative splicing analysis[BarkerSplicing2026], which used rMATS and SUPPA2 on aligned BAM files to identify differential exon usage, intron retention, and alternative 5'/3' splice site events between spaceflight and ground control samples.

Data and code availability

All code and processed data are available at https://github.com/dr-richard-barker/brachypodium-gwas-spaceflight. The interactive results dashboard is available at https://dr-richard-barker.github.io/brachypodium-gwas-spaceflight/. Raw sequencing data are available from NASA OSDR under accession OSD-375 (https://osdr.nasa.gov/bio/repo/data/studies/OSD-375; DOI: 10.26030/2x6b-3v89). SNP data sources are documented in the repository.

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Author contributions

R.B.\ conceived the study, designed and implemented the analysis pipeline, performed gravitropic reorientation assays, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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